MIL-D-36425B
20 June 1978
SUPERSEDING
MIL-D-36425A
29 September 1976

MILITARY SPECIFICATION

DISH, CULTURE, PETRI, TOP AND BOTTOM, COMPLETE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of sterile, disposable, plastic Petri culture dish with a tight lid, for use as a container for a filter disk used in the membrane filtering technique for bacteriological analysis of water under field conditions. The size of the Petri culture dish is commercially referred to as 50 x 9.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, PA. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

3. REQUIREMENTS

* 3.1 Material. The plastic used in fabricating the Petri dish shall be polystyrene. The polystyrene shall be optically clear, and free of toxic lubricants and other additives which would adversely affect the growth of bacterial cultures.

* 3.2 Construction. The Petri dish shall consist of a top section and a bottom section. The design and dimensions of the Petri dish shall be in accordance with Figure 1. The top section (cover) shall consist of a flat, circular top merging into a continuous wall, slightly tapered to fit telescopically with the upstanding wall of the bottom section, forming a tight seal by means of a wedge fit. The bottom section shall have a flat, circular bottom and an upstanding continuous wall, slightly thicker lower part, tapering upward to fit with the top section of the dish. An annular shoulder shall be formed on the lower part of the bottom section to provide a secure closure of the Petri dish. The bottom section shall be provided with radial lugs (nubbins) as indicated in Figure 1. As an alternate, slits (depressions) may be supplied in lieu of the lugs. The bottom section shall be constructed so that it sits flat on a plane and level surface and provides for a uniform media depth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

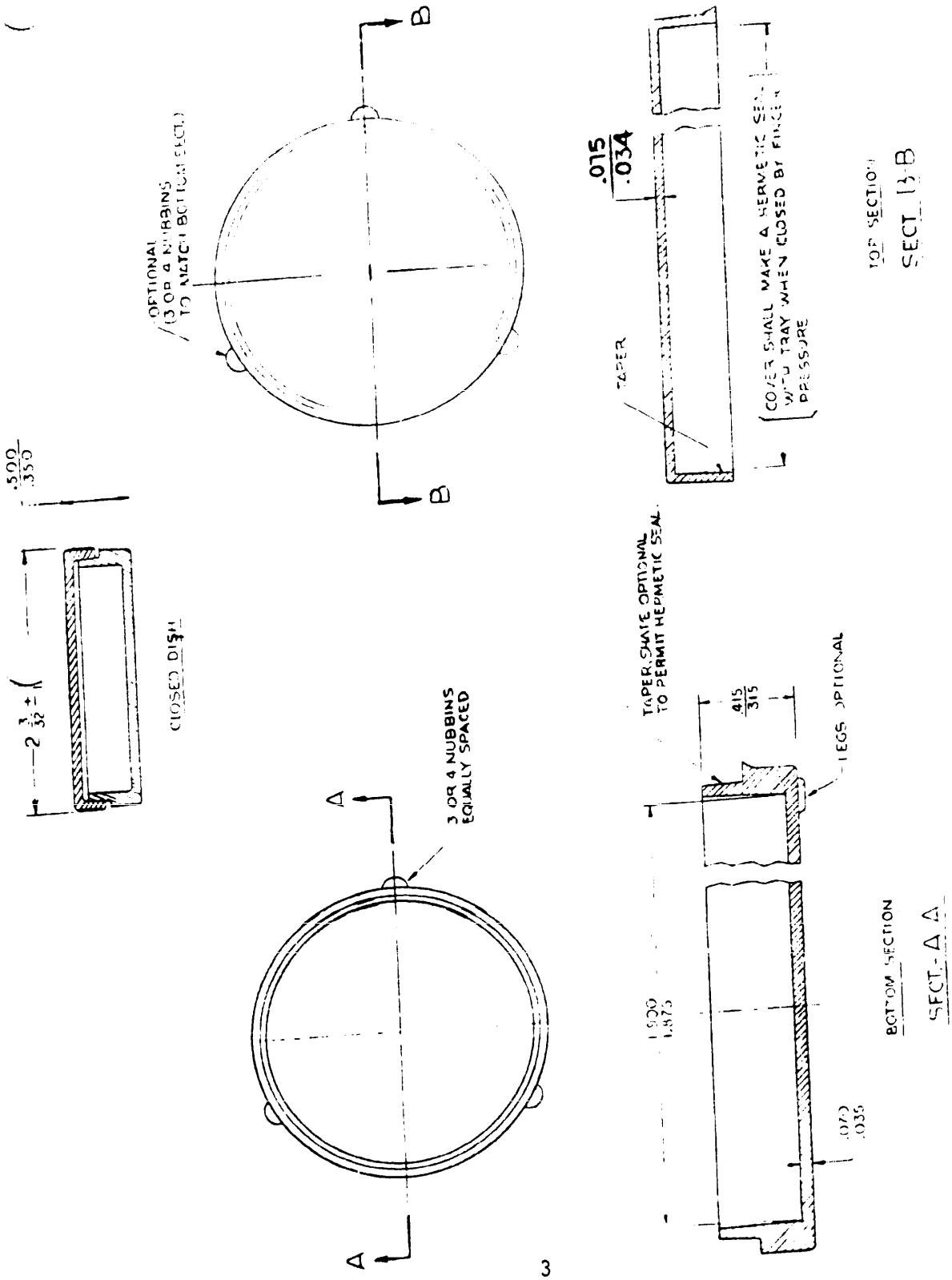


FIGURE 1. Dish, Culture, Petri.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:35Z
Check the source to verify that this is the current version before use.

3.3 Identification marking. The manufacturer's name or registered trademark shall be permanently and legibly marked on the outside top or bottom section. Marking on the top section shall be positioned so as not to interfere with the studies of the cultures.

3.4 Sterility. The inside of the Petri dish shall be sterile, when tested with a biological indicator as specified (see 4.5.1).

3.4.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item Identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip carrier or inoculated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.4.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. Workmanship shall be first class throughout. Each dish shall be free of burrs, sharp edges, loose and other foreign particles, craze, cracks, and other irregularities which detract from its appearance or impair its serviceability. The Petri dish shall be sufficiently scratch-free so as not to interfere with the studies of the culture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:35Z
Check the source to verify that this is the currentversion before use.

TABLE II. For tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
1/	Sterility	3.4	4.5.1	S-1	*

1/ Unless otherwise specified, the product for testing purposes shall be that quantity of Petri dishes that will be supplied in the unit package (see 5.1).

* Acceptance number shall be zero.

4.3 Examination. The dishes shall be examined to determine compliance with all requirements contained in this specification.

* 4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. Dish shall be examined for defects including, but not limited to, those listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Dish not free of cracks.
102	Dish not free of breaks.
103	Dish not free of chips.
104	Dish not free of sharp edges.
105	Petri dish not clean.
106	Petri dish not properly assembled.
107	Petri dish not free of deformation.
108	Petri dish not free of crazing.
109	Petri dish not optically clear.
110	Bottom dish does not sit flat on a plane and level surface.
111	Closure cover cannot effect a tight seal with bottom dish.
Minor	
201	Identification marking not legible, complete and correct.
202	Petri dish not free of burrs and flashes.
203	Petri dish not free of pits, scratches and abrasions which impair visibility.
204	Petri dish not free of unspecified protrusions.

* 4.3.2 Dimensional examination. The Petri dish shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be considered as a defect.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:35Z
Check the source to verify that this is the current version before use.

4.4 Packaging inspection. The inspection of the unit, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Sterility test. The inside of the Petri dish shall be tested for sterility in accordance with the method specified in the USP Procedural Details for Sterility Tests for Non-Parenteral Devices using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.5.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table II.

4.5.1.1 Inoculated product. For each type of sterilization, the organism used shall be as listed below.

Type of Sterilization	Organism	Incubation Temperature
Gas	B. subtilis var niger (or globigii)	30 - 35° C
Radiation	B. pumilus	30 - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.5.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be of equivalent resistance to spore population included in MIL-S-36586. The method of incubation and number of test units shall be as indicated in 4.5.1 (Sterility test) and 4.5.1.1 (Inoculated product). The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.5.1.1.
NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.5.1.3 Alternate method. Suppliers conducting sterility test or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

4.5.1.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing five hundred petri dishes, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Twenty complete dishes (tops and bottoms) shall be packaged sterile in a sealed bag fabricated of cellophane, plastic or other suitable materials. The bag shall be capable of maintaining sterility of contents unless damaged or opened. Twenty-five bags, each containing twenty dishes, shall be packaged in a box conforming to PPP-B-636, class domestic. The box shall be closed as specified in the appendix of the box specification.

5.1.3 Level C. Twenty complete dishes (tops and bottoms) shall be packaged sterile in a sealed bag capable of maintaining sterility of contents unless damaged or opened. Five hundred dishes, packaged twenty dishes to a bag, shall be further packaged in a manner that will afford adequate protection against physical damage from the supply source to the first receiving activity. These packs shall conform to the carriers rules and regulations.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Petri dishes packaged as specified in 5.1.2.1 shall require no further packing except that the unit package shall conform to PPP-B-636, class weather-resistant. In addition, the box shall be waterproofed as specified in the appendix of the box specification.

5.2.2 Level B. Petri dishes packaged as specified in 5.1.2.1 shall require no further packing.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 as specified for exterior containers. The sterility lot or control number shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-9/76-G". Final sterilization shall be on or after indicated packaging date. In addition to any commercial markings, each bag shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, the sterility lot number, quantity contained and the legend or similar statement: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" shall also be shown. The level of protection shall be shown as "A" in

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

lieu of "A/A", "B" in lieu of "A/B" or "C" in lieu of "C/C". Marking shall be printed on the bag or on an insert placed within the bag so as to permit ready identification.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The membrane filter referred to in 1.1 is NSN 6640-00-299-8692, DISK, FILTERING, MICROPOROUS, Hydrosol Type, 47 mm Diameter, 100s. The Petri dish is intended to be used with NSN 6665-00-682-4765, WATER TESTING KIT, BACTERIOLOGICAL.

6.3 This specification does not include all types, classes, grades, sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types normally procured to meet military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-01-030-9012	DISH, CULTURE, PETRI, Top and Bottom, Complete Disposable, 500s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logistics Agency - DM

Project No. 6640-1016

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.

TANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-36425B DISH, CULTURE, PETRI, TOP AND BOTTOM, COMPLETE, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

VENDOR ☐ USER ☐ MANUFACTURER

☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT ? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDED WORDING CHANGE

c. REASON FOR RECOMMENDED CHANGE(S)

REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

FORM 1426
1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:35Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:35Z
Check the source to verify that this is the current version before use.